Topic 14 – Particle model

Students should:		Maths skills
14.1	Use a simple kinetic theory model to explain the different states of matter (solids, liquids and gases) in terms of the movement and arrangement of particles	
14.2	Recall and use the equation: density (kilogram per cubic metre, kg/m^3) = mass (kilogram, kg) ÷ volume (cubic metre, m^3) $\rho = \frac{m}{V}$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d 5c
14.3	*Core Practical: Investigate the densities of solid and liquids*	1a, 1b, 1c, 1d 2a, 2c, 2f 3a, 3b, 3c, 3d 4a, 4c 5c
14.4	Explain the differences in density between the different states of matter in terms of the arrangements of the atoms or molecules	5b
14.5	Describe that when substances melt, freeze, evaporate, boil, condense or sublimate mass is conserved and that these physical changes differ from some chemical changes because the material recovers its original properties if the change is reversed	
14.6	Explain how heating a system will change the energy stored within the system and raise its temperature or produce changes of state	
14.7	Define the terms specific heat capacity and specific latent heat and explain the differences between them	
14.8	Use the equation: change in thermal energy (joule, J) = mass (kilogram, kg) × specific heat capacity (joule per kilogram degree Celsius, J/kg °C) × change in temperature (degree Celsius, °C) $\Delta Q = m \times c \times \Delta\theta$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
14.9	Use the equation: thermal energy for a change of state (joule , J) = mass (kilogram, kg) × specific latent heat (joule per kilogram, J/kg) $Q = m \times L$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
14.10	Explain ways of reducing unwanted energy transfer through thermal insulation	

Students should:		Maths skills
14.11	*Core Practical: Investigate the properties of water by determining the specific heat capacity of water and obtaining a temperature-time graph for melting ice*	1a, 1b, 1c, 1d 2a, 2b, 2f 3a, 3b, 3c, 3d 4a, 4c, 4e
14.12	Explain the pressure of a gas in terms of the motion of its particles	5b
14.13	Explain the effect of changing the temperature of a gas on the velocity of its particles and hence on the pressure produced by a fixed mass of gas at constant volume (qualitative only)	5b
14.14	Describe the term absolute zero, −273 °C, in terms of the lack of movement of particles	
14.15	Convert between the kelvin and Celsius scales	1a 2a
14.16P	Explain that gases can be compressed or expanded by pressure changes	
14.17P	Explain that the pressure of a gas produces a net force at right angles to any surface	
14.18P	Explain the effect of changing the volume of a gas on the rate at which its particles collide with the walls of its container and hence on the pressure produced by a fixed mass of gas at constant temperature	5b
14.19P	Use the equation: $P_1 \times V_1 = P_2 \times V_2$ to calculate pressure or volume for gases of fixed mass at constant temperature	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
14.20P	**Explain why doing work on a gas can increase its temperature, including a bicycle pump**	

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system (1a, 1c, 3c).
- Calculate relevant values of stored energy and energy transfers; convert between newton-metres and joules (1c, 3c).
- Apply the relationship between density, mass and volume to changes where mass is conserved (1a, 1b, 1c, 3c).
- Apply the relationship between change in internal energy of a material and its mass, specific heat capacity and temperature change to calculate the energy change involved; apply the relationship between specific latent heat and mass to calculate the energy change involved in a change of state (1a, 3c, 3d).

Suggested practicals

- Investigate the temperature and volume relationship for a gas.
- Investigate the volume and pressure relationship for a gas.
- Investigate latent heat of vaporisation.